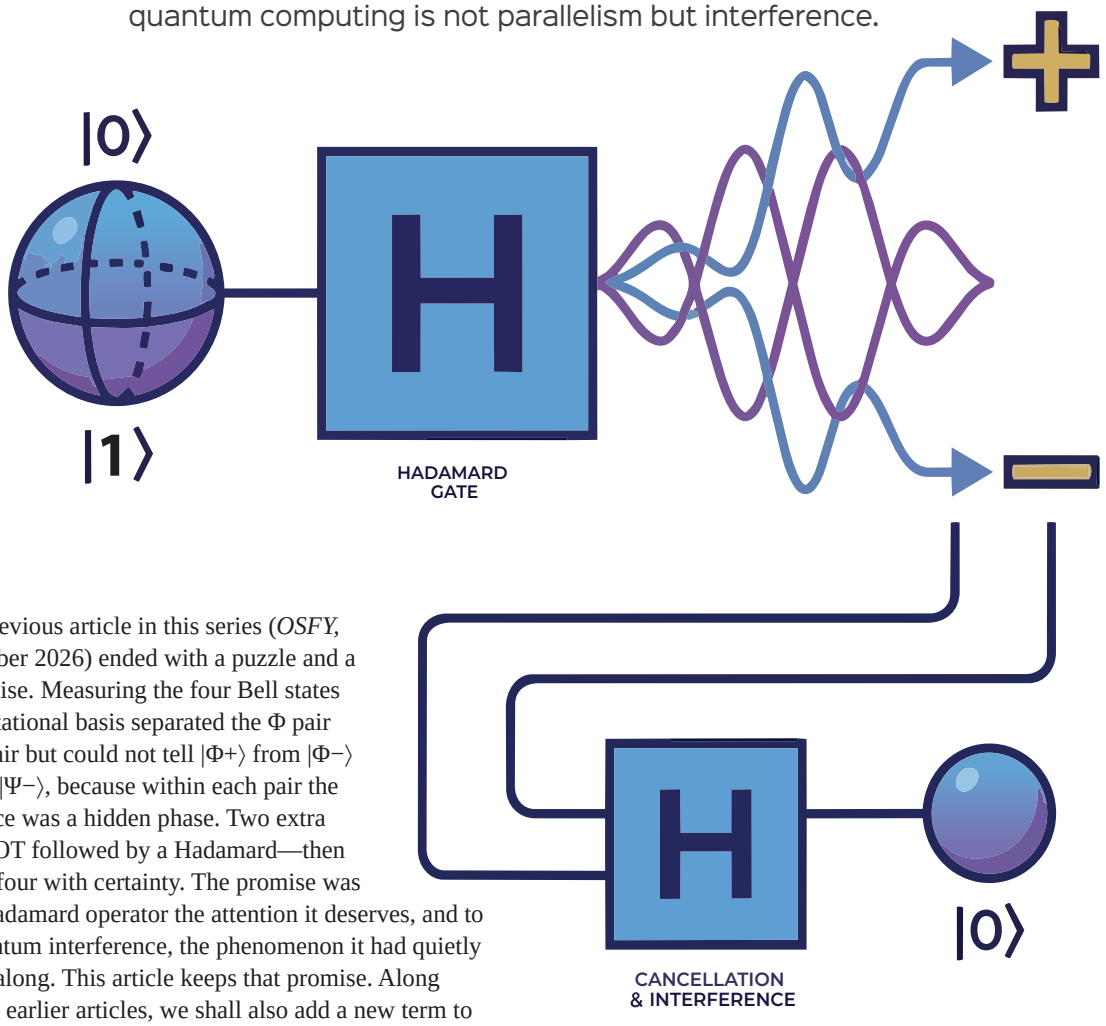
Quantum Computing: Hadamard and the Art of Cancellation

In this sixth article in the series on quantum computing, we take a close look at the Hadamard operator, learn what a basis really is, and discover that the true engine of quantum computing is not parallelism but interference.



The previous article in this series (*OSFY*, October 2026) ended with a puzzle and a promise. Measuring the four Bell states in the computational basis separated the Φ pair from the Ψ pair but could not tell $|\Phi+\rangle$ from $|\Phi-\rangle$ or $|\Psi+\rangle$ from $|\Psi-\rangle$, because within each pair the only difference was a hidden phase. Two extra gates—a CNOT followed by a Hadamard—then identified all four with certainty. The promise was to give the Hadamard operator the attention it deserves, and to examine quantum interference, the phenomenon it had quietly exploited all along. This article keeps that promise. Along the way, as in earlier articles, we shall also add a new term to our quantum computing vocabulary: the no-cloning theorem, which follows naturally from our discussion of bases.

A confession before we begin. The Hadamard gate was the first gate we met in this series and has appeared in almost every program since, yet it has had to wait until the sixth article for a close examination. The official reason for the delay is pedagogical. The unofficial reason is that every equation involving the Hadamard gate carries a $1/\sqrt{2}$, and I have to copy and paste that square root symbol every single time. Jokes apart, the official reason is genuine. The Hadamard gate cannot be fully appreciated without the idea of a basis, a word that has appeared in this series again and again without ever being properly defined. So, being faithful to our three-level approach of code, mathematics and hardware, let us begin with a little linear algebra.

What is a basis? A little linear algebra

Imagine giving directions across India, starting from Nagpur, close to the geographical centre of the country. Any city can